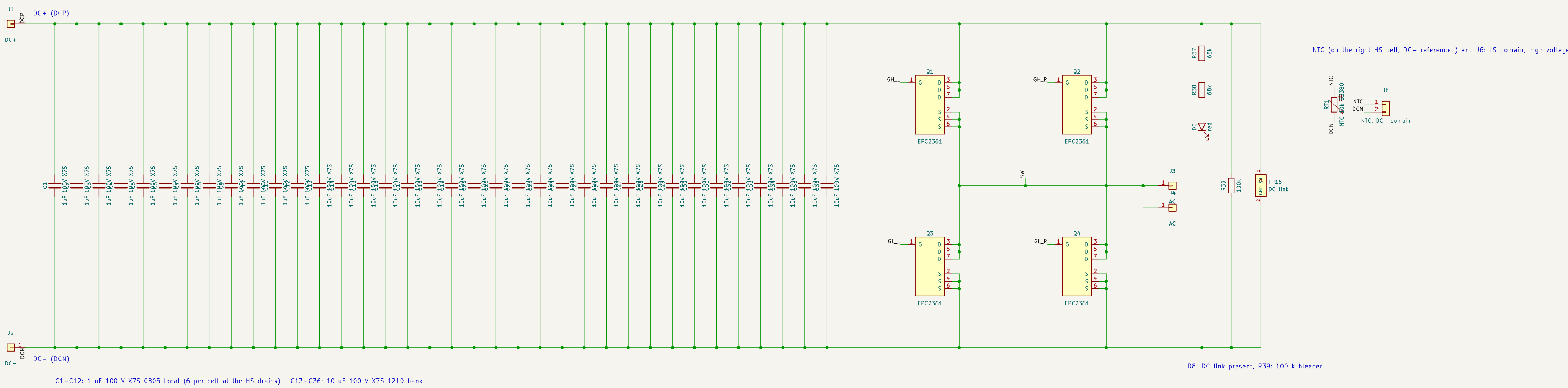
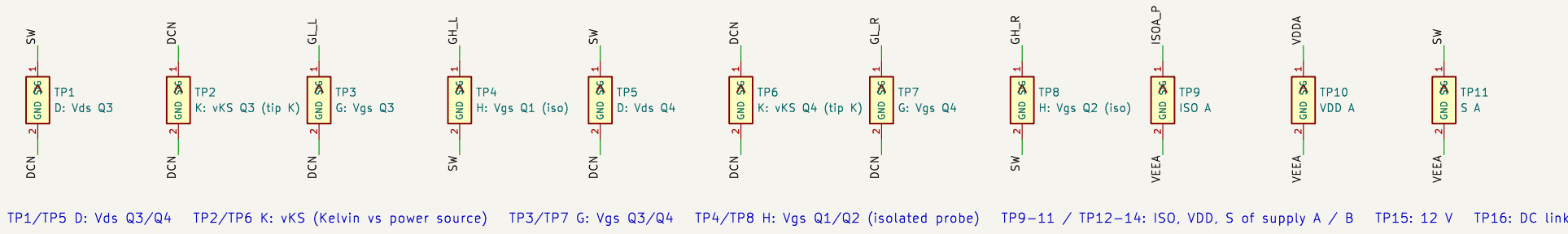


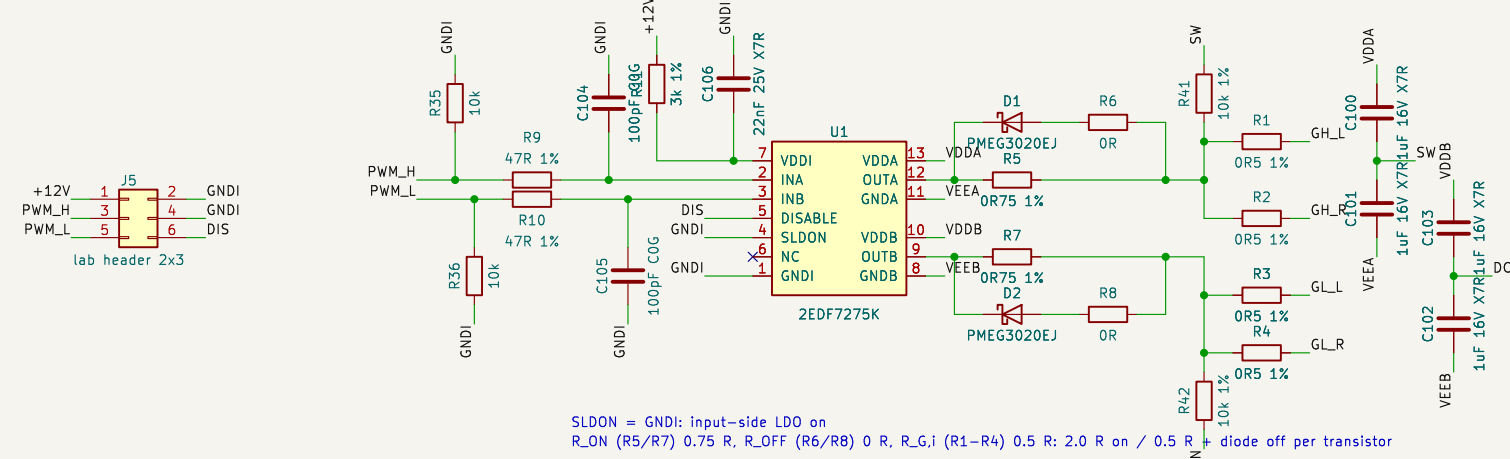
POWER STAGE AND DC LINK (Q1/Q3 left cell, Q2/Q4 right cell; Kelvin = source pad 2 end, see notes)



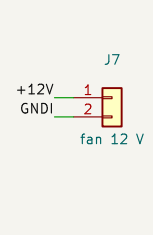
MEASUREMENT LANDINGS (probe pads: tip SIG, spring ground GND at 2.5 / 5.0 mm)



GATE DRIVER U1 (2EDF7275K): input filters, VDDI, separate turn-on / turn-off networks, local decoupling



PRIMARY: fan connector, 12 V LED



MOUNTING: H1-H4 board corners, H5-H8 heatsink (Fischer LAM 4 K 50 12 on an Al spreader, board-only item H51)



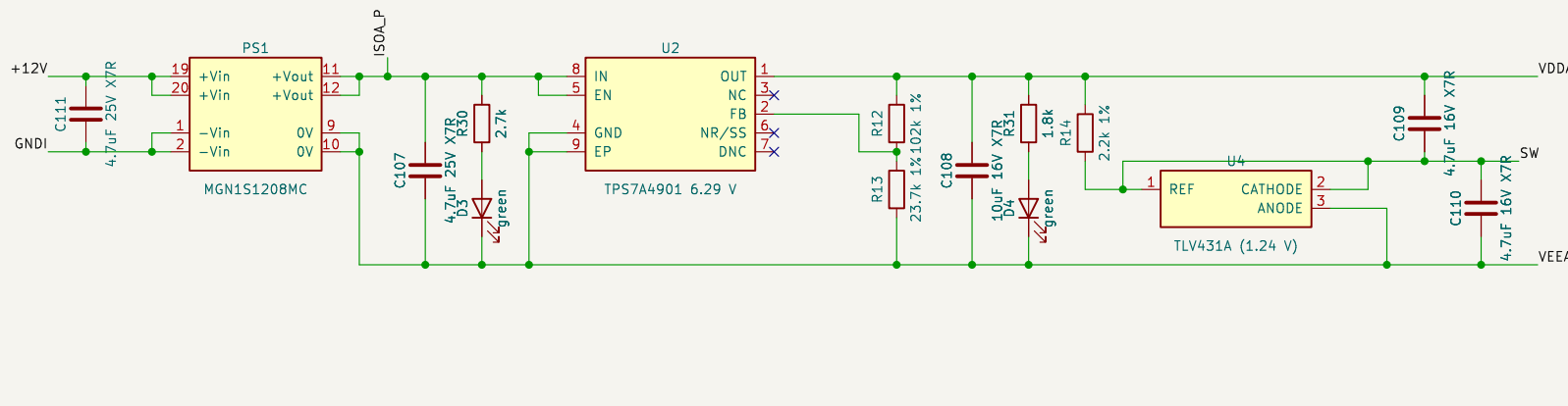
Nets: DCP = DC+, DCN = DC-, SW = switch node. The high-side gate reference S.A is the SW net via the Kelvin traces (source pad 2 of Q1/Q2), the low-side reference S.B is the DCN net via the Kelvin traces of Q3/Q4.

Kelvin routing (PCB): Q1/Q2 pad 2 -> L2 collector -> L4 return under the L3 gate trunk -> C100/C101 node; Q3/Q4 pad 2 -> L2 collector -> C102/C103 node. The schematic shows them as SW / DCN.

U2/U3 TPS7A4901DGNR: EN tied to IN, NR/SS open (CNR = 0 allowed), DNC and NC open, exposed pad to VEE; VOUT = 1.185 V (1 + R_top/R_bot).

TP1-TP16 are probe landings (copper pads, no part). TP2/TP6 measure Kelvin vs power-source copper of the same net (d/dt sensing). J6 and all TPs in supply A/B are in high-voltage domains.

GATE SUPPLY A: MGN1S1208MC 12 V -> 8 V Iso, TPS7A4901 6.29 V, TLV431 sets S = VEE + 1.24 V (S = SW Kelvin)



GATE SUPPLY B: MGN1S1208MC 12 V -> 8 V Iso, TPS7A4901 6.29 V, TLV431 sets S = VEE + 1.24 V (S = DCN Kelvin)

